

Lesson 10 — Design Challenge — Iteration Showcase

Matches LEGO Teacher Portal lesson: “Repeat the Beat”.

Objective (I can...): I can design a machine that uses loops, then iterate it into a measurably better v2.

CS concept: Applying loops (count, forever, nested) with iterative refinement and testing · **Builds on:** Lessons 6–9. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): AI improves by iterating — version after version, each tested. *First-principles question:* “What would you measure to prove your v2 is actually better than your v1, not just different?”

Materials: LEGO CS & AI kit + device per pair (with camera), plan sheets, notebooks. **Model already built** (Loops model) — the build time is plan + code + test, small reconfigs only.

60-minute flow (*As the flex lesson, today's clock trims Build/Code to ~22 to make room for an 8-min AI preview. If a pair needs the full build time, run the preview as a whole-group demo.*) - **Warm-up (8):**

Loop sort — pairs classify example programs as count, forever, or nested, and name a good use for each. (*Theme: Iteration showcase.*) - **Concept + demo (10):** Challenge: build a looped machine (patrol, light show, drummer) that must use at least one loop, then produce a v2 that improves one measurable thing (smoother, faster, more reliable). Coach models choosing a metric. (*Rotate format: “pair teaches pair.”*) -

Build/Code (~22): Pairs plan and code v1 with a loop, pick a metric, then build v2 and record both measurements in the notebook. *Swap at the halfway mark.* - **Meet a real classifier — hands-on**

preview (8): Pairs open **Google Teachable Machine** in the browser and show cards, objects, or hand gestures to the camera using a pre-made or demo model, watching the confidence bars move. **No**

training yet — just see a classifier react live and notice it's never 100% sure. *This previews Unit 6, where they train their own.* **Privacy note, said aloud:** Teachable Machine trains in the browser, on-device; no login is needed and images aren't stored unless someone chooses to save them. - **Share +**

pack-up (5): Showcase — pairs present v1 vs. v2 with their metric and point out the loop; each names one thing the classifier got right or wrong. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator owns the metric and measurements; swap at 15 min. (*Trio rotates plan/code/measure.*)

Differentiation: *Support* — a single-loop target with a provided metric. *Extension / fast-finisher* — require a nested loop and two measured improvements. *Reteach* — reuse a Lesson 7–9 loop as the v1 starting point.

Exit check: Pair delivers a looped build and can show, with a measurement, how v2 improved on v1.

Progress layer: Send home the **Unit 2 parent note** (“your student can use loops and improve a design with evidence”) with the **build photo**. **At-home (optional):** Pick something you do repeatedly; design a v2 of how you do it and name the metric that would show it's better. ~10 min.

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